

Week 6 - Arithmetic and the Eckmann–Hilton argument

1) For this problem, let all variables be integers.

(a) Give a formal definition for $a|b$. (It should start with “ $a|b$ if and only if...”)

(b) Prove or disprove each of the following statements, for $a, b, c \in \mathbb{Z}$.

- If $a|b$ and $a|c$ then $a|bc$.
- If $a|c$ and $b|c$, then $ab|c$.

(c) Prove or disprove each of the following statements, for $a, b, k, n \in \mathbb{Z}$ and $n \geq 1$.

- If $ka \equiv kb \pmod{kn}$, then $a \equiv b \pmod{n}$.
- If $ka \equiv kb \pmod{n}$, then $a \equiv b \pmod{n}$.

2) Give a precise statement of Fermat's little theorem. Suggestion: Include both versions of this theorem you learned in lecture.

3) Find a number $a \in [0, 43)$ such that $44^{43} = a \pmod{43}$.

4) Let $a \in \mathbb{Z}$ be a fixed integer. Prove that there exists an integer $x \in [a - 1, a + 1]$, such that

$$x^{27} \equiv 2 \pmod{3}.$$

5) Find a number $a \in [0, 60)$ such that $2026^{60} \equiv a \pmod{61}$.

The rest of this worksheet will be related to the type of “algebra” from the mock exam (with the $\square$).

6) Let A be a set together with a function $\square : A \times A \rightarrow A$; i.e. for every a, b , we have a distinguished element $a \square b \in A$. Suppose that $\square$ is *associative*, i.e. for all $a, b, c \in A$, we have $(a \square b) \square c = a \square (b \square c)$, and suppose there is a specific *identity* element $e \in A$ such that for all $a \in A$, we have $e \square a = a = a \square e$.

(a) Prove that if $e' \in A$ is an element such that $e' \square a = a = a \square e'$, then actually $e' = e$.

(b) Suppose there is another function $\triangle : A \times A \rightarrow A$ that has the associativity property, and also has an associated identity element, $i \in A$. Prove that if for all $a, b, c, d \in A$, we have

$$(a \square b) \triangle (c \square d) = (a \triangle c) \square (b \triangle d),$$

then actually $\square = \triangle$ and $e = i$. Further, prove that in this situation, the operation $\square = \triangle$ is also *commutative*, that is, $a \square b = b \square a$ (equiv. $a \triangle b = b \triangle a$) for all $a, b \in A$. (The content of this problem is an important tool in my field of research (homotopy theory) called the *Eckmann–Hilton argument*.)